

Leyan(Iris) Lyu

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EDUCATION

University of Maryland, College Park, MD

August 2023 - Expected May 2027

B.S. in Computer Science

SKILLS

Programming: Python, SQL, Java, R, C, MatLab

Data Science & ML: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, MLflow

Data Visualization & Systems & Tools: Plotly, Tableau, Jupyter, AWS, Docker, Linux, Git

EXPERIENCES

Machine Learning Research Lab, University of Maryland (IBM Qiskit & QLab Collaboration)

Machine Learning Research Student

Sep 2025 - Present

- Designed Bayesian neural network models to analyze prediction uncertainty/improve model interpretability.
- Conducted model evaluation and benchmarking experiments across classical/quantum neural network.
- Built automated ML experiment pipelines with **MLflow**, improving experiment reproducibility and tracking efficiency by 30%.
- Performed data analysis/visualized using Python(**Pandas, NumPy, Plotly**) to generate evaluation report.

Machine Learning Research Internship

Jun 2025 - Jul 2025

- Processed and analyzed **665K+** molecular data samples for large-scale machine learning training workflows.
- Implemented distributed training pipelines using **PyTorch Lightning**, improving training efficiency by **20%**.
- Conducted performance benchmarking/ hyperparameter experiments across multiple model configurations.
- Generated quantitative reports to guide model optimization.

University of Maryland, College Park

IT Systems Support Worker

Jul 2025 - Present

- Analyzed system logs and operational metrics to diagnose **100+** Linux/network incidents
- Performed root-cause analysis on system logs and operational metrics.
- Improved documentation workflows for tracking recurring system issues and monitoring system health.

Hupanmodou Foundation, Alibaba Group Female Partners' Foundation

Social Media Technology & Data Analytics Intern

Jun 2024 - Aug 2024

- Analyzed outreach and engagement data to evaluate the effectiveness of digital campaigns.
- Managed website operations including content updates, performance monitoring, and technical maintenance.
- Developed digital content strategies based on audience engagement insights across web and social platforms.
- Produced analytical reports and presentations using Excel/PowerPoint to support program decision-making.

PROJECT

AI Pet Co-Regulation & Emotional Reflection Platform

Feb 2026 - Present

- Designed and implemented a **scalable, real-time** multi-turn conversational system simulating pet personas, improving user engagement through personalized interaction experiences
- Developed backend APIs (Python) for request processing, dialogue state management and storage
- Integrated LLM APIs into a **production-style inference pipeline** for real-time response generation with contextual awareness
- Developed prompt engineering frameworks (persona modeling, tone control, context injection) to ensure **consistency and personalization at scale**
- Implemented **logging and analytics pipelines** to monitor system performance and enable **data-driven optimization** of response quality